Michael A. Yong

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SUMMARY

AI-focused software engineer who builds Python automations, LLM-powered tools and dashboards that save business teams hours of manual work. Builds secure AI assistants with Claude, MCP and MySQL, and works hands-on across the Microsoft 365 stack (Power BI, Entra ID, Intune).

SKILLS

AI & Machine Learning: Claude (Agent SDK), Model Context Protocol (MCP), prompt engineering, ChatGPT, Microsoft Copilot, PyTorch, computer vision

Languages: Python, SQL, JavaScript, C#, Java

Data: MySQL, Pandas, NumPy, Power BI, Tableau, Excel (macros), APIs

Microsoft & Cloud: Microsoft 365, Entra ID, Intune, Azure Blob Storage, AWS (EC2, S3)

Web & Tools: FastAPI, React, Vite, Node.js, JWT, Git/GitHub, Jupyter, pytest

EXPERIENCE

Nova Consulting | *Engineering Intern, Miami, FL* May 2024 – Jan 2025

- Built API connections between the BI platform and back-end data servers, giving the team direct, automated access to source data and streamlining how reports were prepared across multiple projects.
- Automated manual data-processing steps into a semi-automated macro workflow, saving the team 8–10 hours every week and freeing analysts to focus on higher-value reporting and analysis.
- Connected Python scripts to LLM tools such as ChatGPT and Microsoft Copilot to automate recurring data-analysis tasks, speeding up team analytics and improving collaboration across teams.
- Designed and deployed Power BI dashboards that gave management self-serve visibility into operational data, shortening meeting preparation and supporting faster, data-driven decisions.
- Secured the Microsoft 365 environment by configuring Entra ID Conditional Access and Intune compliance policies for company and BYOD devices, reducing unauthorized access incidents.

Threshold 360, Inc. | *Software Engineer Intern, Lakeland, FL* Oct 2023 – Apr 2024

- Led development of a computer-vision object detection model that powers features in the company's virtual tour platform, helping the product recognize key objects within each space.
- Created an image-processing method that accurately measures door widths from tour imagery, adding accessibility data that shows users how easily a space can be navigated.

PROJECTS

AI Data Assistant | *Python, FastAPI, Claude Agent SDK, MCP, MySQL, React, Azure Blob Storage* Nov 2026 – Present

- Built an AI analyst assistant that answers plain-English questions about a fictional infrastructure investment portfolio, using Claude through the Agent SDK behind a React chat interface.
- Developed a Python MCP server giving Claude read-only MySQL access, combining SELECT-only query validation with restricted database users so the AI cannot change or delete data.
- Added document upload to Azure Blob Storage, developed locally with Azurite, with Claude summaries that cite the source file and page for every claim, plus one-page company summaries.
- Secured the app with JWT sign-in, role-based access enforced by database permissions, rate limiting, and prompt injection tests confirming hidden instructions in documents are ignored.
- Wrote a pipeline that cleans messy portfolio data before loading MySQL, and logged every question, the tools Claude used and response time to support auditing and usage reporting.

EDUCATION

Florida Polytechnic University | *B.S. in Computer Science* 2024